

Preferential Accretion onto Eccentric and Unequal Binary Black Hole Systems

Stanislav DeLaurentiis^{1,2,4} , Zoltan Haiman^{2,3} , Magdalena Siwek^{2,4} , Roman Rafikov^{1,3,5} 

¹Department of Astronomy, Columbia University, 550 W. 120th Street, New York, NY 10027, USA

²Department of Physics, Columbia University, 550 W. 120th Street, New York, NY 10027, USA

³Institute of Science and Technology Austria (ISTA), Am Campus 1, 3400 Klosterneuburg, Austria

⁴Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Wilberforce Road, Cambridge CB3 0WA, UK

⁵Institute for Advanced Study, Einstein Drive, Princeton, NJ 08540, USA

ABSTRACT

Supermassive Binary Black Holes (SMBBHs) are expected to be surrounded by Circumbinary Disks (CBDs) which affect the binary through gravitational torques and accretion. It has been reported that the binary can experience “preferential accretion” where one black hole (BH) out-accretes the other for hundreds of orbits, but the phenomenon has yet to be fully described or understood. In this work, we utilize a suite of 80 SMBBH hydrodynamic simulations with varying mass ratios (q_b) and eccentricity (e_b) in order to robustly delineate the behavior of preferential accretion, determine its relationship to the structure of the CBD, and study its observational consequences. In this paper we report the most extensive series of measurements for $\dot{q}$, the rate of change of the binary mass-ratio, and $\lambda(t)$ the ratio of the BH accretion rates rate-of-change $\dot{q} \equiv d/dt(M_2/M_1)$ and the accretion-rate ratio $\lambda(t) \equiv \dot{M}_2(t)/\dot{M}_1(t)$, both as functions of q_b and e_b . We also share our findings that a) the accretion behavior onto one BH over the other is strongly tied to time-variability of the accretion-rate ratio onto the two BHs tracks the precession of the CBD, b) there can be sub and super-eddington accretion in a single binary, and c) there exists a regime where the CBD can drive the binary away from $q_b = 1$ binary evolves toward $q_b \neq 1$ during the gas-dominated phase. Given these findings, we suggest flickering jets and an under-density of LISA events $q_b = 1$ a relative dearth of strictly $q_b = 1$ LISA events as potential observable consequences.

1 INTRODUCTION

Cosmic structure forms hierarchically and galaxy mergers are expected to result in gravitationally bound Super-Massive Binary Black Holes (SMBBHs) (White & Rees 1978; Begelman et al. 1980) (White & Rees 1978; Begelman et al. 1980); the same disc-mediated dynamics also operate in binary stars and protoplanetary systems. During galactic merger, it is also expected that the inter-stellar medium from the proto-galaxies are funneled to the galactic center (Barnes & Hernquist 1992). This creates a reservoir of material that, due to the conservation of angular momentum and the binary’s gravitational potential, forms a circumbinary disk (CBD). Many aspects of the CBD have been well studied in the literature, in both the SMBBH and the star-planet contexts. It has been shown that the binary can influence the CBD, creating eccentric structure and precessing eigenmodes, (Lubow 1991; Whitehurst 1994; Nelson 2003; Goodchild & Ogilvie 2006; Lubow 1991; Whitehurst 1994; Nelson 2003; Goodchild & Ogilvie 2006) and that the CBD can in turn influence the binary—changing both its semi-major axis and eccentricity (Tiede et al. 2020; Zrake et al. 2021; D’Orazio et al. 2013; Siwek et al. 2023b; D’Orazio & Duffell 2021; Farris et al. 2014), as well as exciting precession of the binary itself (Tiede et al. 2024; Dittmann et al. 2023; Calcino et al. 2023). However, crucially, the binary also accretes from the circumbinary disk, influencing both the mass-ratio and associated light-curves of the system (e.g. Siwek et al. (2023a); Westernacher-Schneider et al. (2022)).

It has been a widely known result of CBD studies that the

accretion on to the binary ($\dot{M}_b = \dot{M}_1 + \dot{M}_2$) is not steady but experiences variability through time. Near equal-mass ratio ($q_b > 0.7$) circular binaries display a sawtooth pattern with a period of about 5 binary orbital periods (τ_b) potentially due to the presence of an $m = 1$ over-density traveling at the inner edge of the CBD (MacFadyen & Milosavljević 2008; Muñoz et al. 2019; Westernacher-Schneider et al. 2022; Muñoz et al. 2019; Westernacher-Schneider et al. 2022). Eccentric and unequal mass-ratio binaries too display variability on the order of a binary period (Muñoz et al. 2019; Westernacher-Schneider et al. 2022). In order to study how the binary, as an entity, accreted material from the circumbinary disk, Tiede et al. (2022) tracked the paths of passive tracer particles in a two-dimensional grid-based hydrodynamical codes. They determined that much of the gas is viscously transported from the outer disk before adopting a ballistic trajectory and ultimately accreting on to the BHs near the inner edge. They also delineated an accretion horizon of the binary semi-major axis of the binary (a_b), a radius that past which any material enters is accreted, inside of which any gas that enters is captured by one of the BHs.

However, studies have also pointed out the discrepancy between the accretion behaviors of the primary and secondary components of the SMBBH (e.g. Siwek et al. (2023a); D’Orazio et al. (2024) Miranda et al. (2017); Siwek et al. (2023a); D’Orazio et al. (2024)). Namely, the secondary tends to accrete, on average, at a greater rate than the primary in what has become known as “preferential accretion”—pushing the system to equal mass ratio ($q_b = 1$)

(~~Siwek et al. 2023a~~)(Miranda et al. 2017; Siwek et al. 2023a).

While systems with $q_b = 1$ generally accrete at equal rates, suggesting a stable equilibrium at $q_b = 1$, studies have reported that at certain eccentricities (e.g., $e_b = 0.5$ and $e_b = 0.6$), the binary can undergo transient "symmetry breaking." (see Figure 7 in Muñoz et al. (2019)). In these instances, one black hole temporarily accretes more than its counterpart before its accretion rate is suppressed, while the other experiences an enhanced accretion rate.

This process was explored in Siwek et al. (2023a) by studying the accretion rate behavior across 50 2D hydrodynamic simulations of SMBBHs in CBDs, sampling different binary parameters q_b and e_b . Siwek et al. (2023a) reported time averaged values of $\lambda = \dot{M}_2/\dot{M}_1$, the ratio of the secondary's accretion rate to the primary's, for their simulation suite and displayed snapshots of the corresponding CBD. Siwek et al. (2023a) report that low-eccentricity binaries display larger values of λ than high-eccentricity binaries ($e_b > 0.6$). Further, the process of the flip in preferential accretion was suggested to be associated with the behavior of the CBD. Namely, it was associated with a unique disk behavior they called "forced precession". DeLaurentiis et al. (2025) also discussed the preferential accretion rate of the binary, suggesting it be understood in geometric terms though the distance from the BHs to the CBD.

The protoplanetary disk community have run two and three dimensional smoothed particle hydrodynamic (SPH) simulations of CBDs in interest of understanding the accretion onto the primary and secondary (Günther & Kley 2002; Ochi et al. 2005; Young et al. 2015). ~~In To date, in~~ simulations run for ~~for~~ $\leq 100\tau_b$, ~~they~~, ~~they have~~ found that the primary can out-accrete the secondary due to streamlines that, despite entering the Lagrange point L2, have such high angular momentum that they are able to flow to the counterpart (Ochi et al. 2005; Young et al. 2015). Similar findings were suggested in Tiede et al. (2022).

Despite these efforts we are still yet to fully characterize and understand the rich detail of accretion onto the individual components of a binary in a CBD system. Thus, in the following work we build on Siwek et al. (2023a) and, with the consent of the authors, use the larger simulation suite of Siwek et al. (2023b) to report the largest binary parameter sweep study on "preferential accretion". Further, we more robustly characterize the accretion behavior by applying unique numerical techniques and more fine-grained analytic tools.

In Section 2 we briefly discuss the technical details of the simulations, the key concepts from linear theory we draw upon, and the key techniques we employ. In Section 3 we present our findings: detailing the temporal behavior of preferential accretion and mass-ratio evolution and describe how that varies with initial binary q_b and e_b . In Section 4 we detail the consequences our findings have on observations. We discuss how our results could suggest a) jets that periodically switch on and off, "flickering", and a population of SMBBHs with $q_b \neq 1$. In Section 5 we summarize our key findings and discuss next steps.

2 ANALYTIC ~~TOOLS~~ ~~TOOLS~~ AND NUMERICAL ~~METHODS~~ ~~NUMERICAL METHODS~~

In the following section we briefly describe the setup of the Siwek et al. (2023a,b) simulations, the concepts from linear disk theory we employed, and the numerical tools we leveraged.

2.1 Simulation ~~Setup~~ ~~setup~~

In the following we briefly describe the setup for the simulations of interest, ~~but please see Siwek et al. (2023a,b) and refer readers to Siwek et al. (2023a,b)~~ for a more thorough discussion.

Siwek et al. (2023a,b) performed 2D hydrodynamical simulations of binary black holes embedded in a finite, locally isothermal disk with an α -viscosity of $\alpha = 0.1$ and an aspect ratio of $h = 0.1$. The disk is initialized ~~according to a power-law function that places the~~ ~~with a power-law surface-density profile and a corresponding temperature profile (set by the locally isothermal condition);~~ the inner edge of the cavity sits at $2a_b$ and the outer edge at $\approx 50a_b \approx 50a_b$. The binary-disk system spans a computational grid of $300a_b \times 300a_b$ with open boundary conditions, allowing the disk to ~~viscously spread over time and reach settle into~~ a quasi-steady state.

The binary is modeled as two sink particles with a mass-ratio q_b and radii $r_s = 0.03a_b$, moving on a fixed Keplerian orbit of eccentricity e_b . The fraction of gas accreted from each gas cell at each time step by the sink particle is $\gamma_0 \left(1 - \frac{r_{ij}}{r_s}\right)^2$ where r_{ij} is the radial distance from the j th sink particle to the i th gas cell. In addition to the mass of the gas cell, the sink also accretes the gas' linear momentum.

The simulations, conducted with the moving-mesh code AREPO (Springel 2010), used Voronoi tessellation to generate a grid of cells and explored a wide parameter space for q_b and e_b across the sets $q_b \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ and $e_b \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8\}$. The simulations were run for 10,000 orbits, with surface density snapshots recorded at ~~aepocenter~~ every 10 orbits. Accretion rates, however, were recorded independently at a much higher cadence, with a time-step of ≈ 0.01 orbits.

2.2 Disk ~~Theory~~ ~~theory~~

In order to illuminate the relationship between the accretion of the binary and the behavior of the CBD we must first characterize the time-variable behavior and attributes of the disk. Namely, we focus on the eccentricity and precession of the disk.

Gas eccentricity within CBDs is expected to grow through mechanisms such as eccentric Lindblad resonances (ELRs) or spiral shock pumping at the cavity edge (~~Lubow 1991; Whitehurst 1994; Paardekooper et al. 2008; Kley et al. 2008; Shi et al. 2012; Lubow 1991; Whitehurst 1994; Paardekooper et al. 2008; Kley et al. 2008; ?~~). While ELRs and spiral shocks promote eccentricity growth (~~Lubow 1991; Shi et al. 2012~~)(Lubow 1991; ?), viscous damping (Goodchild & Ogilvie 2006) and orbit crossings (Statler 2001) act to suppress it. Previous 2D simulations have found steady-state eccentricity profiles, indicating that these competing effects can reach equilibrium (Miranda et al. 2017; Siwek et al. 2023a). Further, 2D and 3D hydrodynamical simulations have found significant eccentricity near the inner edge of the disk (the cavity), with eccentricity declining outward (MacFadyen & Milosavljević 2008; Miranda et al. 2017; Siwek et al. 2023a; Ragusa et al. 2024). Similar trends have been seen in magneto-hydrodynamic simulations (~~Shi et al. 2012~~)(?), suggesting that eccentricity is a robust characteristic of CBDs.

Additionally, simulations have demonstrated that the CBD can precess (Nelson 2003; Shi et al. 2012; Miranda et al. 2017; Thun et al. 2017; Siwek et al. 2023a; Nelson 2003; ?; Miranda et al. 2017; Thun et al. 2017; Siwek et al. 2023a), with precession frequencies often attributed to the eigenmodes of a Schrödinger-like equation for eccentricity evolution

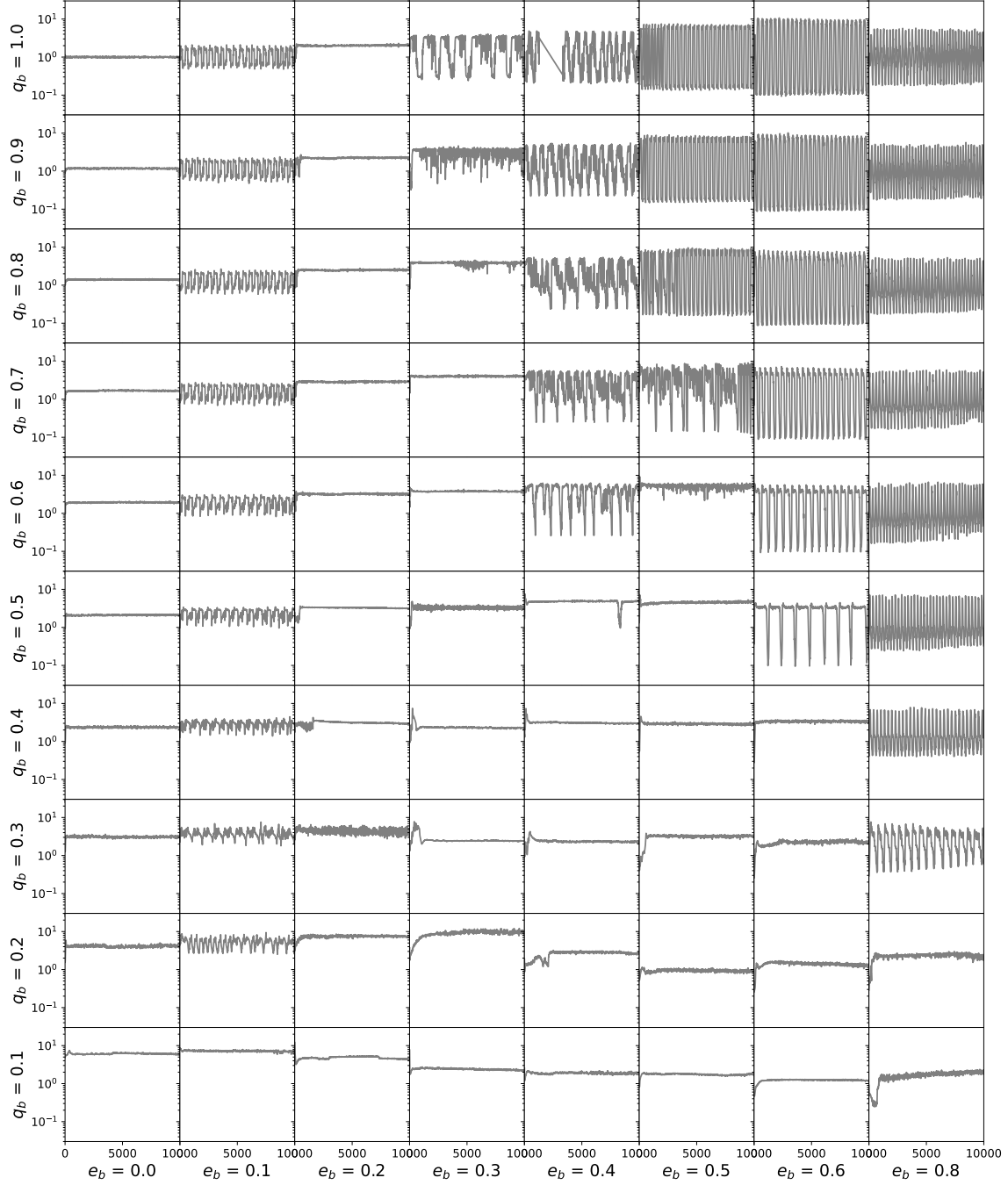


Figure 1. $\lambda(t)$ —The accretion-rate ratio $\lambda(t) \equiv \dot{M}_2(t)/\dot{M}_1(t)$ for our binary simulation—the entire 80-simulation suite. Each panel represents a unique (e_b, q_b) simulation, with $\lambda(t)$ on the y-axis and time on the x-axis (in units of binary orbital periods τ_b). The panel's position on the larger grid signifies its binary parameters, with e_b being constant along columns and increasing from left to right, and q_b being constant along rows and increasing from bottom to top.

<u>1.0</u>	P	P	L	P	P	P	P	P
<u>0.9</u>	P	P	L	L	P	P	P	P
<u>0.8</u>	P	P	L	L	P	P	P	P
<u>0.7</u>	P	P	L	L	P	P	P	P
<u>0.6</u>	P	P	L	L	P	L	P	P
<u>0.5</u>	P	P	L	L	L	L	P	P
<u>0.4</u>	P	P	L	L	L	L	L	P
<u>0.3</u>	P	P	L	L	L	L	L	P
<u>0.2</u>	P	P	L	L	L	L	L	L
<u>0.1</u>	P	L	L	L	L	L	L	L
q_b/e_b	<u>0.0</u>	<u>0.1</u>	<u>0.2</u>	<u>0.3</u>	<u>0.4</u>	<u>0.5</u>	<u>0.6</u>	<u>0.8</u>

Table 2. Reproduction of Table 1 from DeLaurentiis & Rafikov (2025), showing whether the CBD about a binary of given q_b and e_b is locked (L; red) or precessing (P; blue). The CBD behavior is determined by evaluating the time-series of the eccentricity vector at $a \leq 5a_b$.

have a precessing CBD—suggesting that the preferential accretion of the binary corresponds to a locked CBD — suggesting that the time-variability of preferential accretion is, in some part, determined paced by the behavior of the CBD and thereby the cavity. Siwek et al. (2023a) described the disk as exhibiting three distinct regimes: This is a statement about time-variability, complementary to the time-averaged framework of Siwek et al. (2023a), who classified disks into three regimes (free precession, forced precession, or a locked disk). Through symmetry arguments, Siwek et al. (2023a) suggested that freely precessing or locked disks corresponded to a locked) and applied symmetry arguments to predict the time-averaged λ of unity, while disks undergoing forced precession show preferential accretion (λ). The reader is referred to their Section 3.5 for a discussion of the symmetry-breaking and “preferential accretion switching” for the forced-precession regime.

The $e_b = 0$ simulations break this one-to-one mapping. Though $e_b = 0$ simulations have been well documented (Siwek et al. 2023a) to have precessing disks, it is quite clear that their preferential accretion is time-stable. Due to the one-to-one mapping for non-circular binaries, this $e_b = 0$ simulations break our time-variability mapping but not the time-averaged framework: their CBDs precess freely yet their $\lambda(t)$ is constant. This suggests that a binary may need to have an apocenter in order for the CBD behavior to influence its accretion. This is somewhat a non-zero apocenter for CBD precession to imprint itself on the time-variability of accretion, similar to the argument made in DeLaurentiis et al. (2025) which suggested that there needed to be a minimum eccentricity in precessing binary simulations to see is required for strong modulations in $\lambda(t)$.

Further, we note that the amplitude of $\lambda(t)$ oscillations are not uniform among time-varying simulations. In fact, we note that there seems to be a relationship is a clear correlation between the amplitude of the $\lambda(t)$ oscillation and the e_b of the binary.

In Fig. 2 we display a heat-map indicating the variability of $\lambda(t)$. The cell values are colored according to the standard deviation of

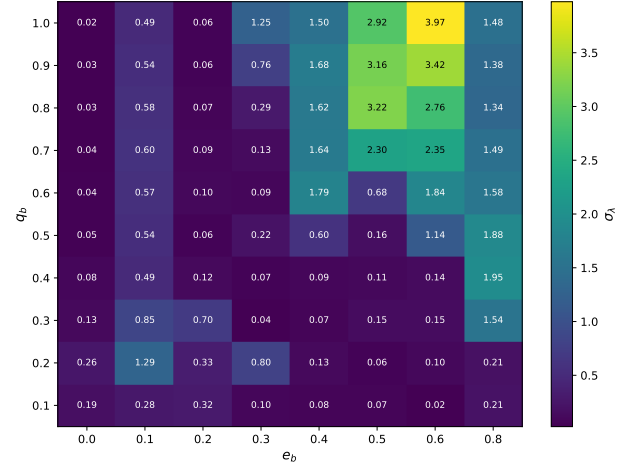


Figure 2. A heat-map displaying Heat-map of σ_λ , the standard deviation of $\lambda(t)$ to indicate over the difference post-instability ($t > 3000\tau_b$) portion of the simulation. Larger values (brighter) indicate stronger time-variability of preferential accretion. The variability peaks broadly in oscillation-behavior across our simulations the high- e_b , intermediate- q_b region of parameter space.

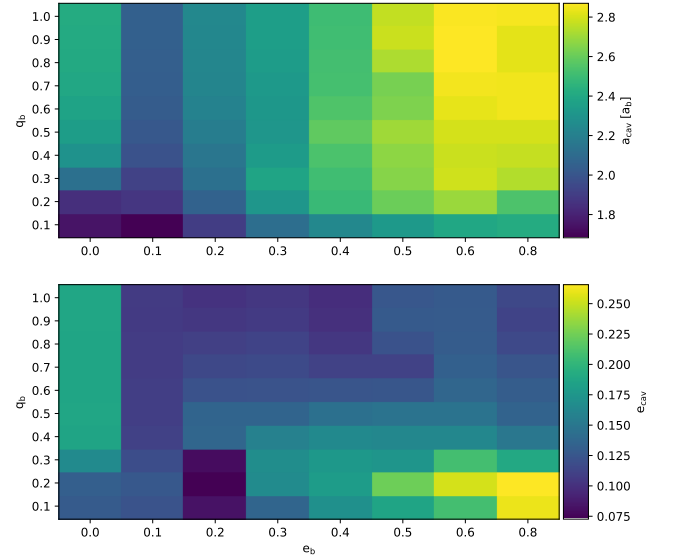


Figure 3. Reproduction of Figure 18 from DeLaurentiis & Rafikov (2025) displaying the semi-major axis and eccentricity of the cavities of our simulations with varying binary eccentricity and mass ratio. The cavity is characterized by the $m = 1$ Fourier mode of the CBD cavity

$\lambda(t)^3 - \sigma_\lambda$. We note that while many of our simulations have small σ_λ since they are time-stable (see Table 1), those that display mean-

³ For our calculations of the behavior of $\lambda(t)$ we make a cut at 3000 orbits, discard the first $3000\tau_b$ of each time-series (corresponding to not let the initial time-instability of disk-instability transient documented in DeLaurentiis & Rafikov 2025). The remaining $7000\tau_b$ is sampled at the disk (DeLaurentiis & Rafikov 2025) influence our metrics snapshot cadence of $10\tau_b$, giving $N = 700$ usable points per simulation.

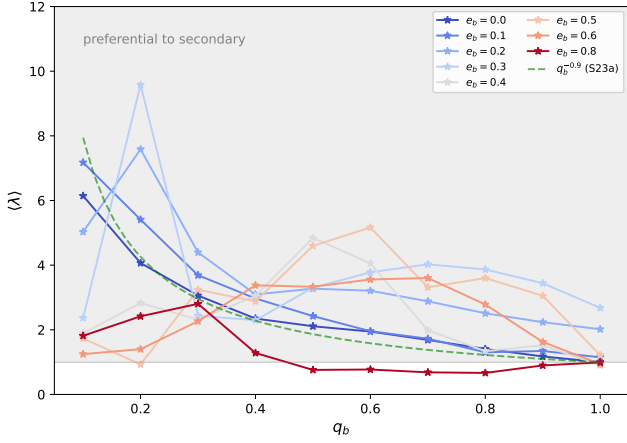


Figure 4. $\langle \lambda \rangle$ across Time-averaged accretion-rate ratio $\langle \lambda \rangle = \langle \dot{M}_2 / \dot{M}_1 \rangle$ as a function of q_b (x-axis) and e_b (line color). $\langle \lambda \rangle$ is represented on the y-axis. The gray-shaded region marks $\langle \lambda \rangle > 1$, q_b is on the x-axis, and the color represents e_b simulation labelled “secondary” accretes preferentially. We also plot the power-law fitting to $\langle \lambda \rangle$ of $q_b^{-0.9}$ from Siwek et al. (2023a) in the dashed green line is the $q_b^{-0.9}$ power-law fit reported by Siwek et al. (2023a) for circular binaries.

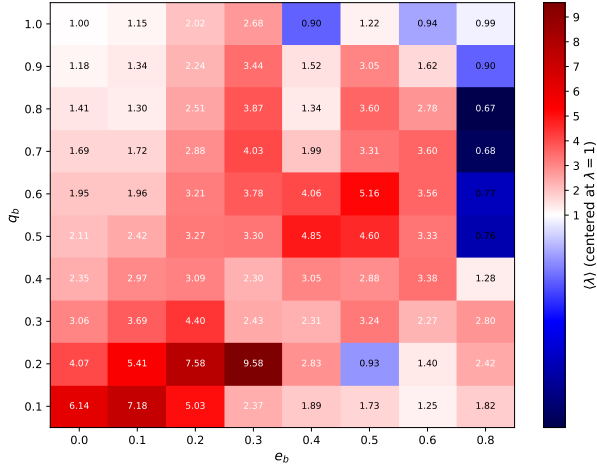


Figure 5. A heat-map displaying the time-averaged Heat-map of $\lambda(t)$ across our simulation suite. The diverging colormap is centered at $\lambda = 1$: red cells indicate preferential accretion onto the (apocenter-defined) secondary BH, blue cells indicate preferential accretion onto the primary. More eccentric, more unequal mass binaries seem tend to accrete more preferentially, though the trend is not monotonic.

ingful σ_λ suggest a modest trend. Namely, we find that σ_λ seems to increase with e_b , peaking at $e_b = 0.6$. This modest increase in σ_λ with e_b is reminiscent of Fig. 3, Figure 18 of DeLaurentiis & Rafikov (2025), which indicates that the eccentricity of the cavity increases with e_b and peaks at $e_b = 0.6$. This correlation further suggests that the CBD is a crucial regulator of preferential accretion.

In addition to temporal nature of $\lambda(t)$ we also comment on the mean value of $\lambda(t)$, determining which BH is preferred to accrete and to what extent. We report our results in Fig. 4 and Fig. 5.

In Fig. 4 we display the mean values of $\lambda(t)$ on the y-axis and the

q_b on the x-axis, while the color represents the e_b . We find that our figure is in broad agreement with Figure 4 of Siwek et al. (2023a). We too find that the $\langle \lambda \rangle$ of the $e_b = 0$ simulation seems to follow the power law of $q_b^{-0.9}$. However, we also find that $e_b = 0.1$ also holds true to this line, suggesting that the behavior of $\langle \lambda \rangle$ is generic to lower eccentricity binaries and not unique to $e_b = 0$. Further, we note that all our unequal-mass simulations suggest that if there is preferential accretion it is always to the secondary BH. This is in line with all the results of the literature (Muñoz et al. 2019; Duffell et al. 2020; Farris et al. 2014; Dittmann & Ryan 2021; Siwek et al. 2023a). Despite preferential accretion being biased towards the secondary, we note that the extent to which one BH accretes over the other depends greatly on the binary parameters. The behavior of $\langle \lambda \rangle$ is perhaps best shown in Fig. 5.

Fig. 5 display q_b on the y-axis and e_b on the x-axis, plotting the value of $\langle \lambda \rangle$ according to color. On first glance, it is clear that some binary parameters lend themselves to stronger preferential accretion rates ($\langle \lambda \rangle$) than others. Namely, in Fig. 5 we find that low e_b and low q_b values display high preferential accretion values, while high e_b and high q_b display lower ones as discussed in Siwek et al. (2023a). This is the opposite of the trend we saw for σ_λ values in Fig. 2 and for the eccentricity of the CBD (DeLaurentiis & Rafikov 2025). This suggests that the binary parameters themselves are more important than the CBD in regulating the level of preferential accretion.

While there is certainly a trend, the level of preferential accretion is by no means monotonic with respect to q_b and e_b . We note that intermediate values (e.g. $e_b, q_b = (0.4, 0.4)$, $(0.5, 0.5)$ e.g. $(e_b, q_b) = (0.4, 0.4)$ and $(0.5, 0.5)$) have higher preferential accretion levels than their neighbors. Further, Fig. 5 displays interesting “hotspots” of high preferential accretion, with $e_b = 0.3$ $q_b = 0.2$ $(e_b, q_b) = (0.3, 0.2)$ being the simulation with the highest levels of preferential accretion. We also find curious dimspots where the accretion onto both BHs is near-equal despite the binary having unequal mass. Notably, $q_b = 0.2$ $e_b = 0.5$ has a $(e_b, q_b) = (0.5, 0.2)$ has $\langle \lambda \rangle = 0.93$.

More importantly, we would like to note a point of deviation from Siwek et al. (2023a). In Fig. 4 and Fig. 5 we report $\langle \lambda \rangle \neq 1$ for $e_b, q_b = (0.2, 1.0)$, $(0.3, 1.0)$ the $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ simulations. The $e_b = 0.2$, $q_b = 1.0$ $(e_b, q_b) = (0.2, 1.0)$ simulation in Siwek et al. (2023a) had a numerical error, resulting in a deviation in λ . Accounting for this, we determine that for $e_b, q_b = (0.2, 1.0)$, $(0.3, 1.0)$ the $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ simulations, the binary, in fact, preferentially accretes onto one of the BHs. The implications of this are further discussed in Section 3.2 and Section 4.

Clearly, there is a variability in both the time behavior of preferential accretion and its value across simulations. In the following section we make a preliminary attempt to understand preferential accretion as an effect of the unique geometry between the binary and the CBD.

3.1.1 Preliminary Analysis

In Section 3.1 we noted similarities between the preferential accretion behavior $\lambda(t)$ and the CBD geometry. The delineation of time-stable and time-varying $\lambda(t)$ in Table 1 was likened to the precessing and locked circumbinary disks. σ_λ seemed to display the same trend with e_b and q_b as that of the CBD eccentricity. While alternative explanations have been proposed (Artymowicz 1983), the assumption that the preferential accretion of the binary is related to the relative closeness of the components to the CBD’s inner edge (Rafikov 2016; Farris et al. 2014) has been left unchallenged. In the following

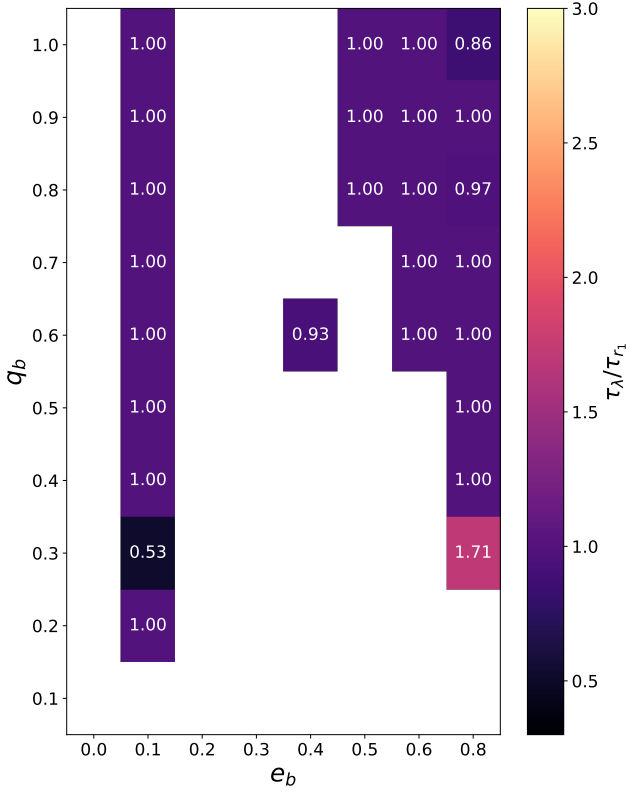


Figure 6. The ratio of the period of preferential accretion (τ_λ) to the period of the CBD precession, τ_{r_1} . The x-axis is the eccentricity of the binary, the y-axis is the mass-ratio of the binary, and the color of the cells correspond to the ratio of the periods. The uncolored cells are those where a period for r_1 or λ could not be determined.

section we provide a first, preliminary, analysis of this assumption and alternative ways in which the CBD would influence preference accretion.

As discussed in Section 2, we use the dominant Fourier modes of the cavity to reconstruct its shape as a function of θ : $r_{\text{cav}}(\theta)$. Combined with the position of the BHs, we then define the distance from the inner edge of the CBD, “the cavity wall”, to each BH as a function of θ . We take θ_1^* and θ_2^* such that $r_{\text{cav}}(\theta)$ is minimized for each BH. We then calculate $r_1 = r_{\text{cav}}(\theta_1^*) - r_{\text{BH}_1}$ and $r_2 = r_{\text{cav}}(\theta_2^*) - r_{\text{BH}_2}$ where r_1 and r_2 are the closest distances from the cavity-wall to the primary and secondary BH, respectively (Fig. 7). Since our BH positions are fixed—as we only record snapshots at apocenter— r_1 and r_2 simply encode the information about the size and orientation of the CBD and serve as proxies.

A first step in determining the relationship between preferential accretion and the CBD is to study the temporal behavior. We have already determined that, except for the $e_b = 0$ case, all precessing CBDs result in a non-time stable $\lambda(t)$. $r_1(t)$ and $r_2(t)$, proxies for the CBD orientation, display the same split into time-stable and time-varying behavior as $\lambda(t)$ in Table 1. For simulations with time-varying $\lambda(t)$ we compare their period of oscillation with that of r_1 and r_2 . To determine the period of each time-series we first calculate a fast Fourier transform (FFT) and normalize the amplitudes by their sum. The period is defined at that of the largest amplitude in the spectrum which has a normalized value > 0.05 . We find the periods

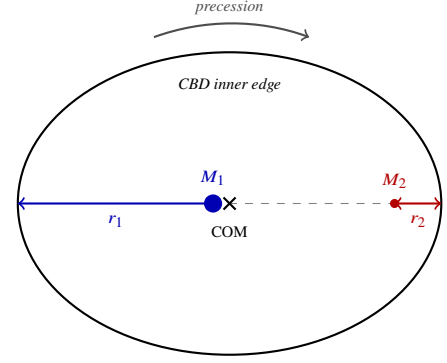


Figure 7. Schematic of the cavity geometry at apocenter for the locked-CBD case, illustrated for an unequal-mass binary with $q_b \ll 1$. The CBD inner edge (black ellipse) is closer to the secondary BH (M_2 , red) on one side; the primary (M_1 , blue) is farther from its nearby cavity wall. Closest distances from each BH to the wall are r_1 and r_2 , with $r_1 > r_2$ in the locked configuration shown. As the disk precesses (curved gray arrow above the ellipse), the ellipse’s major axis rotates, and the orientations of r_1 and r_2 change accordingly.

of r_1 ⁴ and λ in this fashion and display the ratio of the two periods in Fig. 6.

We note that the ratios reported in Fig. 6 are nearly all equal to one. Though certain binaries have ratios that deviate from unity we note that upon further investigation, these deviations do not suggest resonances between CBD precession and preferential accretion but rather point to the difficulty of FFT to handle FFT being unstable when applied to periodic, non-sinusoidal $\lambda(t)$ for such simulations (e.g. $e_b = 0.4$ (e.g. $e_b = 0.4$, $q_b = 0.8$). Thus, we find that the precession of the CBD determines not only whether and the variability of $\lambda(t)$ varies but also the period at which it varies; are tightly correlated in both occurrence and period: we observe the same FFT period in both quantities and the same locked-vs-precessing partition.

We pause to address a methodological subtlety. Our snapshots are recorded only at apocenter, every 10 binary orbital periods. This cadence is fast enough to resolve the cavity precession across our suite, since the precession period is always much longer than 20 orbits. Each snapshot therefore catches the cavity at a slightly rotated orientation, and the time-series $r_1(t)$ and $r_2(t)$ inherit the cavity’s precession frequency. Two empirical checks confirm this. First, in every simulation r_1 and r_2 share the same FFT period — which is exactly what we expect if both are sampling the same precessing cavity from different sides. Second, that shared period matches the period of $\lambda(t)$ (Fig. 6). Fig. 8 makes both checks visible: for $(e_b, q_b) = (0.8, 0.3)$ the three signals each peak cleanly at $\tau \approx 700\tau_b$ above our 0.05 amplitude threshold, while (0.4, 0.7) shows the same period concentration but at lower amplitudes — the regime in which the FFT becomes unstable when applied to non-sinusoidal $\lambda(t)$. The period match between $\lambda(t)$ and $r_1(t)$, $r_2(t)$ therefore licenses the inference that $\lambda(t)$ variability is paced by the cavity’s apsidal precession, even though the precession itself is not directly resolved at our snapshot cadence.

In Fig. 9 we display r_1 (blue) r_2 (red) and λ (black) for a slice in time for six of our simulations, with the dual y-axis representing the dimensionless lambda value (left) and the distance from each BH to the edge of the cavity in units of a (right).

⁴ It is important to note that we have found the period of r_1 and r_2 to be exactly equal in all our simulations, making either appropriate for this calculation.

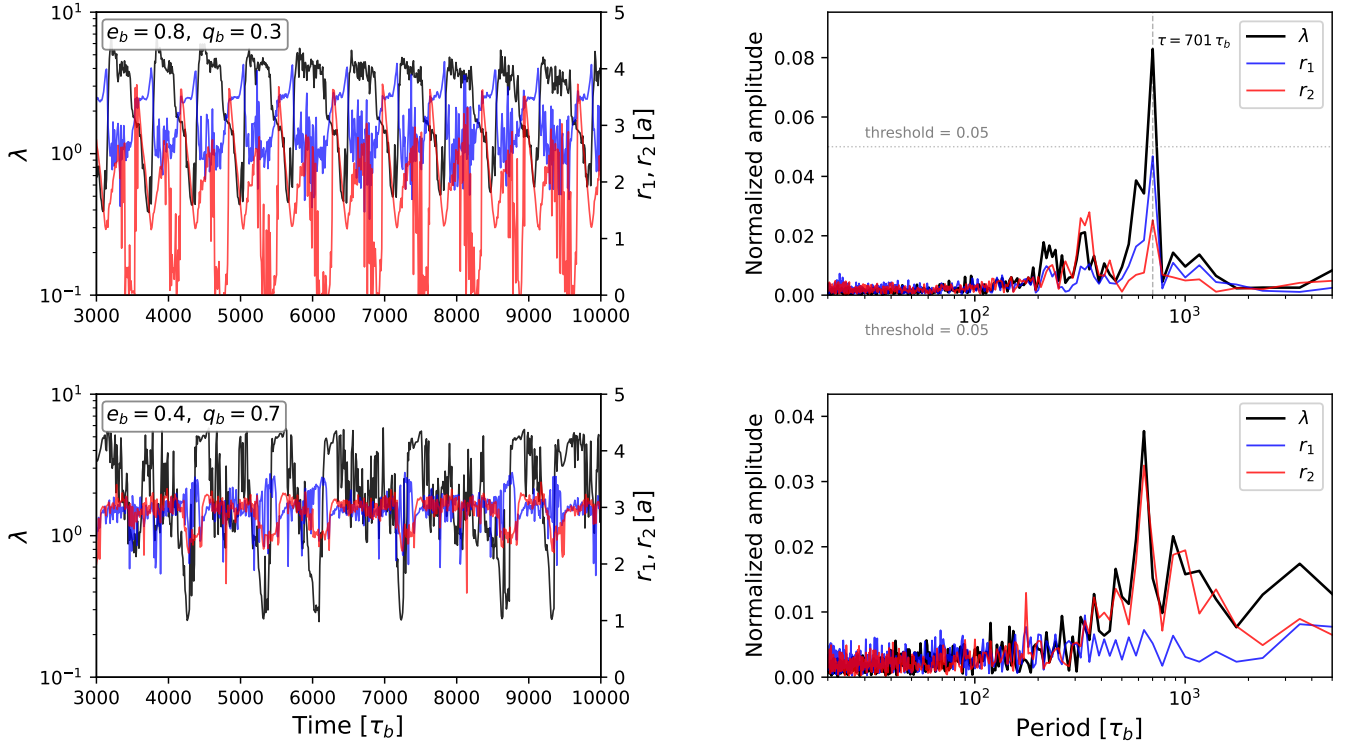


Figure 8. Time-series and power spectra for the two messy cases of Fig. 9: $(e_b, q_b) = (0.8, 0.3)$ (top row) and $(0.4, 0.7)$ (bottom row). Left column: the same $\lambda(t)$ (black, left log-axis), $r_1(t)$ (blue), and $r_2(t)$ (red, right axis in units of a) shown over the post-transient window $3000 \leq t/\tau_b \leq 10000$ ($N = 701$ samples at the snapshot cadence of $10\tau_b$). Right column: normalized rFFT amplitudes $|\tilde{X}(f)|/|\Sigma_f \tilde{X}(f)|$ as a function of period $\tau = 1/f$; the dotted line marks the 0.05 amplitude threshold used in Fig. 6. For $(0.8, 0.3)$ all three signals peak at $\tau \approx 700\tau_b$, with λ 's peak well above threshold; for $(0.4, 0.7)$ the peak sits in the same period range but below threshold, illustrating why the period ratio in Fig. 6 departs from unity for non-sinusoidal $\lambda(t)$.

As indicated by Fig. 6 the variability of the distance from the BHs to the cavity displays the same period (or lack thereof) as λ . This is abundantly clear for the simulations in the left column of Fig. 9, however we note that it is also true for some of the simulations in the right column where the period is time-varying. It is especially well captured by $e_b = 0.5, q_b = 0.7$ where the change from a stochastic to periodic λ signal at $\approx 8500\tau$ is mirrored in the change of behavior of r_1 , and r_2 . Further, in $e_b = 0.4, q_b = 0.7$ we find that large amplitude changes in r_1 and r_2 (e.g. $\approx 6000, 9000\tau$) occur simultaneously with large amplitude changes in λ .

The naive understanding of the tie between preferential accretion and the distance to the black holes is as follows. The inner cavity rim hosts the densest, slowest-moving-slowest-moving gas in the system ([Farris et al. 2014](#); [Duffell et al. 2020](#)) ([Farris et al. 2014](#); [Duffell et al. 2020](#)). Whenever a black hole's Hill sphere overlaps that rim, the steep local gravitational-potential gradient bends incoming streamlines into its mini-disk, and the instantaneous capture rate obeys a Bondi-Hoyle-like scaling:

$$\dot{M}_{\text{near}} \propto \Sigma_{\text{rim}} v_{\text{rel}}^{-3} \quad \text{Bondi-Hoyle-like scaling}$$

$$\dot{M}_{\text{near}} \propto \Sigma_{\text{rim}} v_{\text{rel}}^{-3}. \quad (4)$$

A companion just-a-few- R_{Hill} farther from the rim encounters lower

surface density and higher gas-BH relative velocity ([??](#)), so-

$$\frac{\dot{M}_{\text{far}}}{\dot{M}_{\text{near}}} \sim \frac{\Sigma_{\text{rim}}}{\Sigma_{\text{rim}}} \left(\frac{v_{\text{rel}}}{v_{\text{rel, near}}} \right)^3 \ll 1,$$

gas-BH relative velocity ([Miranda et al. 2017](#); [D'Orazio & Duffell 2021](#)) , so

$$\frac{\dot{M}_{\text{far}}}{\dot{M}_{\text{near}}} \sim \frac{\Sigma_{\text{rim}}}{\Sigma_{\text{rim}}} \left(\frac{v_{\text{rel}}}{v_{\text{rel, near}}} \right)^3 \ll 1, \quad (5)$$

suppressing its accretion by orders of magnitude even though both black holes share the same global gas reservoir.

In $e_b = 0.2, q_b = 0.1$ ($(e_b, q_b) = (0.2, 0.1)$) we see what we would expect from our naive explanation. The panel [MSS: is this still figure 6?] corresponding panel of Fig. 9 shows that this simulation has a CBD that is locked such that our primary black hole (blue) is about twice as far from the cavity edge than our secondary black hole (red). Correspondingly, we see that the secondary is accreting at a rate about five times the primary, seemingly as a result of the secondary's greater pull on its neighboring portion of the CBD. In fact, we can even take this further and note the fact that the tidal potential-on-the-gas-by-the-secondary-is-about-field of the secondary on its nearby cavity wall is approximately four to five times greater-than-its-companion, the same factor difference as the stronger than the tidal field of the primary on its more distant cavity wall (with the tidal field scaling as M/r^3 and $r_1/r_2 \approx 3.4$ measured from the simulation snapshot), in agreement with the observed factor of ~ 5 in the relative accretion rate. However, our naive approxima-

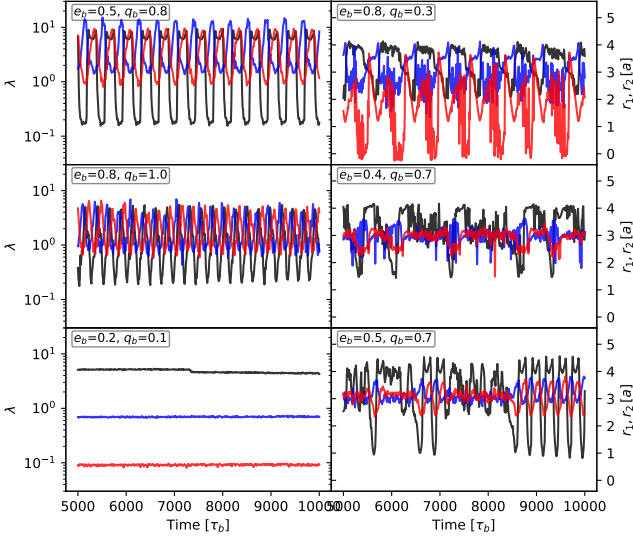


Figure 9. Comparison between the time-series of $\lambda(t)$ (black, left log-axis), $r_1(t)$ (blue), and $r_2(t)$ (red) (right axis, in units of the binary semi-major axis a). Time is in binary orbital periods τ_b ; panels are plotted for $5000 \leq t/\tau_b \leq 10000$. Clockwise from upper-left we display $e_b, q_b = (0.5, 0.8)$, $(0.8, 0.3)$, $(0.8, 1.0)$, $(0.4, 0.7)$, $(0.2, 0.1)$, and $(0.5, 0.7)$. We find that the BHs' position with respect to the cavity does not determine the components' relative accretion rates.

tion is inherently over-simplistic. It doesn't take into account each BH's minidisk, the streams by which the gas is fed to the BH, or any non-linear fluid dynamics, such as shocks, that may be relevant in this mechanism. Studying the other panels, we see that our naive explanation no longer suffices.

Before discussing the deviations, we make the naive prediction explicit. If preferential accretion were governed solely by proximity to the cavity wall, then for a binary at apocenter with the cavity oriented toward the secondary, r_2 should be at its minimum (cavity wall closest to secondary) precisely when λ is at its maximum (secondary accreting most). As the cavity precesses, r_2 should rise to its maximum a half-precession-period later, when λ should be at its minimum. We therefore expect $r_2(t)$ to be exactly π out of phase with $\lambda(t)$, while $r_1(t)$ — by symmetry, since the cavity wall is then closest to the primary — should be exactly in phase with $\lambda(t)$.

All simulations displayed in Fig. 9, except for $e_b = 0.2, q_b = 0.1$ ($e_b, q_b = (0.2, 0.1)$), show that lower r_2 values do not correspond to higher accretion rates onto the secondary. The naive explanation suggests that r_1 should be exactly “in-phase” with λ while r_2 should be exactly $\pi/2$ “out-of-phase”. However, we find that $e_b, q_b = (0.5, 0.8)$ and $e_b, q_b = (0.8, 1.0)$ both simulations ($e_b, q_b = (0.5, 0.8)$ and $(0.8, 1.0)$) clearly do not display such behavior. The former displays that the naive behavior: the former shows both r_1 and r_2 are at $\pi/4$ out of phase with λ , while the latter displays shows the exact opposite to our naive expectation, that r_1 is in fact “in-phase” with r_1 in phase and r_2 is “out-of-phase” out of phase with λ . Further, when turning to $e_b, q_b = (0.8, 0.3)$ and $e_b, q_b = (0.4, 0.7)$ we note that our naive approximation further breaks down, as now ($e_b, q_b = (0.8, 0.3)$ and $(0.4, 0.7)$) break the naive picture in another way: the amplitude of r_1 relative to r_2 does not strictly coincide with one BH having preferred accretion over the other. We note that for $e_b, q_b = (0.8, 0.3)$ which

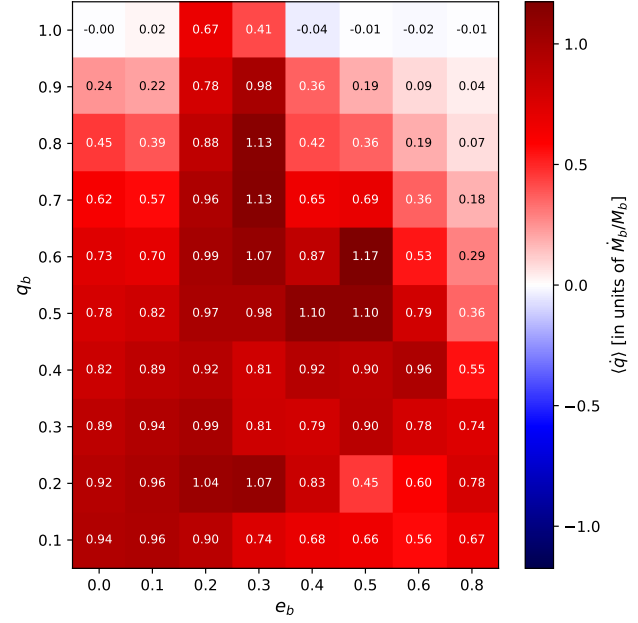


Figure 10. The average time-averaged rate of change of the binary mass-ratio, $\langle \dot{q} \rangle$, for our simulation suite, in M_b/M_b units. The x-axis is the binary eccentricity e_b and the y-axis is the binary mass-ratio q_b . The diverging colormap is centered at $\langle \dot{q} \rangle = 0$: red cells indicate the value of binary evolves toward $q = 1$ (positive $\langle \dot{q} \rangle$), blue cells away from $q = 1$. Most $q_b < 1$ cells are positive (driven toward equal mass); the $q_b = 1$ row is represented as color mostly consistent with zero, with notable exceptions at $e_b = 0.2$ and 0.3 where the binary evolves away from unity (§3.2).

BH accretes preferentially. For $(0.8, 0.3)$, while the primary is always at least as far away from the cavity wall from the cavity wall as the secondary, we still see periods where the primary accretes preferentially. Further, for $e_b, q_b = (0.4, 0.7)$ we see a similar issue in that preferentially. For $(0.4, 0.7)$, during periods where both the primary and secondary BHs are equidistant from the cavity wall, the secondary is accreting accretes up to 4 times more than the primary⁵.

From the above analysis it is clear that that the naive explanation of distance to the cavity and tidal potential field is not robust. While the variability and periods of both are linked, they are far from causal $\lambda(t)$ and r_1, r_2 are correlated, but with no clear causal link between them. While this study has focused on describing the zoo plethora of preferential accretion behaviors and its ties to the CBD, a more robust explanation on how the binary preferentially accretes will be explored in a forthcoming paper.

3.2 Mass Ratioratio

In addition to preferential accretion, we also report the average rate of change of the mass ratio, $\langle \dot{q} \rangle$. Namely, we calculate $\dot{q}(t)$ as described in equation 1 in Section 2, make a time-cut at 3000τ to ensure that early instabilities do not effect our results, and report the mean on the truncated time-series in units of $\dot{M}_b/M_b$ in Fig. 10.

⁵ While λ is a ratio and this behavior could be due to small amplitude deviations in $\dot{M}_1$ and $\dot{M}_2$, we found that the amplitude of the sum is not strongly suppressed during this period.

What first strikes us in Fig. 10 is that for all $q_b \neq 1$ simulations the binary $\langle \dot{q} \rangle$ is positive, meaning that the binary is evolving towards equal mass. We note, however, that the rate of this evolution is quite varied, spanning two orders of magnitude. Further, the gradient of $\langle \dot{q} \rangle$ in parameter space is not uniform. In fact, we see that there exists a massive drop off in $\langle \dot{q} \rangle$ in the upper right corner of Fig. 10, starting along the diagonal extending from $q_b = 0.8$, $e_b = 0.6$.

Turning to the upper row of Fig. 10 we find that while most of the $q_b = 1$ simulations display near zero values of $\langle \dot{q} \rangle$ ⁶ we find that there are two simulations which notably do not. Namely, the $q_b = 1$, $e_b = 0.2$ and $q_b = 1$, $e_b = 0.3$ simulations⁷, the result is best framed as the presence or absence of mass-ratio evolution among initially equal-mass binaries: most $q_b = 1$ simulations display $\langle \dot{q} \rangle$ values consistent with zero⁶ (absence of evolution), while $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ both display strong positive $\langle \dot{q} \rangle$ values, meaning that the “secondary”⁷ BH is growing more massive than the primary. However, since this a (presence of evolution away from unity).

We pause to clarify a notation subtlety. Our convention $q_b \equiv M_2/M_1 \leq 1$ assigns “primary” to the more massive BH and “secondary” to the less massive one. In a strictly $q_b = 1$ case, this non-zero $\langle \dot{q} \rangle$ means that the binary is in fact being fed in such a way that it is evolving system this assignment is degenerate, and we adopt the convention — following Siwek et al. (2023a) — of identifying the components by their spatial location at apocenter. Once $q_b = 1$ is broken by accretion, the BH initially labeled “secondary” grows into the more massive component, formally inverting the $q_b \leq 1$ convention. We continue to label the BHs by their initial assignment throughout the integration for clarity. The statement that the binary “evolves away from unity” should be read as: the mass ratio M_2/M_1 deviates from unity, where the labels 1 and 2 refer to the original assignment, even after the inversion.

This result for $e_b, q_b = (0.2, 1.0)$ ($e_b, q_b) = (0.2, 1.0)$ stands as a correction to Siwek et al. (2023a), where it was reported that q_b would remain at one-unity for this case. However, after ensuring to correctly track the sink particles in the $q_b = 1$ simulations After ensuring correct sink-particle tracking in the $q_b = 1$ simulations, we find that accretion towards unity for SMBBHs is not necessarily a given. This makes sense especially in light of the results of DeLaurentiis & Rafikov (2025), where it was reported that the $e_b, q_b = (0.2, 1.0)$ and $(0.3, 1.0)$ ($e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ simulations have stable, locked disks in line with Siwek et al. (2023a), with a stable non-varying λ , which in this case was coincident with the primary being further from the cavity edge than the secondary.

The finding that equal-mass binaries at $e_b = 0.2, 0.3$ will $e_b = 0.2, 0.3$ accrete away from equal mass is an entirely novel result and has deep implications on has implications for CBD structure and SMBBH population statistics. Firstly, from From DeLaurentiis & Rafikov (2025) we note both $e_b, q_b = (0.2, 1.0)$ and $(0.2, 0.9)$ that both $(e_b, q_b) = (0.2, 1.0)$ and $(0.2, 0.9)$ simulations have locked disks in roughly the same orientation, with the pericenter of the disk being closest towards closest to the secondary. However, we note that

⁶ The negative values reported are within the standard error of the mean for the $\dot{q}$ time-series ($\approx 10^{-2}$ for $q_b = 1$) and can thus be taken to be zero.

⁷ The slightly negative values are within the standard error of the mean for the $\dot{q}$ time-series ($\sigma \approx 10^{-2}$ for $q_b = 1$, estimated from the variance of $\dot{q}(t)$ divided by the integration duration), and can thus be taken to be zero.

⁸ In the equal-mass case, “secondary” and “primary” merely identify the components of the binary which at apocenter are in the same spatial quadrant as the secondary and primary of an unequal-mass binary.

As the $q_b = 1$ case is accreting-accretes away from unity in such a way that the BH which was initially, the BH initially identified as the “secondary” becomes the primary. This would likely cause grows into the primary; the disk, which is oriented towards the secondary, to realign itself—flipping according to oriented toward the original secondary, must therefore realign itself, flipping in concert with the switch in primary and secondary position. The details of identities. We speculate that this realignment proceeds on the disk’s reaction to this change in binary parameters would not only provide key insights into the mechanism behind CBD orientation, but could provide insight not only apsidal precession timescale, since the same precession dynamics that orient locked disks in the first place are the natural mechanism by which a locked disk can re-orient. The details of this reaction would provide insight into the CBD-orientation mechanism, into how far away from unity the binary evolves, but ultimately evolves, and — depending on the process of the geometry and timescale of re-orientation could prove to be — could constitute an event with characteristic EM signatures. Confirming this picture would require live-binary simulations through a sustained $\dot{q} \neq 0$ phase, which we leave to future work.

4 OBSERVATIONAL IMPLICATIONS

In the following section we discuss the observational consequences of our $\lambda(t)$ and $\dot{q}$ results (see Section 3).

4.1 Flickering Jets

A key observational consequence of accretion onto BHs is the potential launching of jets. While the exact mechanism to launch jets continues to be an extremely active area of study, we can broadly understand jet launching through the lens of the jet launching can be broadly understood through the radiative efficiency of the accretion disk. When an accretion the disk is radiatively efficient the photons, and thus, photons (and the energy they provide, carry) are efficiently radiated away from the disk. However, in radiatively inefficient disk, the photons get trapped in the disk, causing a buildup of energy and pressure, both. In radiatively inefficient disks, photons are trapped, building up thermal and magnetic pressure. This scenario can result in produce large pressure gradients that could drive jets, as well as a strong buildup in magnetic field strength which could capable of driving jets, and a strong magnetic field configuration whose energy can be extracted via the Blandford-Znajek-Blandford-Znajek mechanism. Radiatively inefficient, geometrically thick disks are commonly found at low accretion rates $\dot{M} < 0.01 \dot{M}_{\text{Edd}}$ ($\dot{M} < 0.01 \dot{M}_{\text{Edd}}$, (Guolo et al. 2021) and at high accretion rates $\dot{M} > \dot{M}_{\text{Edd}}$ ($\dot{M} > \dot{M}_{\text{Edd}}$). In both regimes, the geometric thickness of the disk plays a second role beyond setting the radiative efficiency: the inflated inner walls form a funnel along the BH spin axis that channels magnetic flux and outgoing material into a collimated relativistic jet. Given these criteria for jet launching, we are able to make statements about the jets from our simulations.

To do so, we must first scale our numerical accretion rates, which are in units M_{bin}/τ , to Eddington units. To be able to retain information about the relative accretion rates of each BH to the other, we must set the accretion rate of the binary $\dot{M}_b \equiv \dot{M}_1 + \dot{M}_2$ to be $\gamma \dot{M}_{\text{Edd}}$ where γ is an arbitrary scale factor and The Eddington accretion rate $\dot{M}_{\text{Edd}}$ is the steady accretion rate at which radiation pressure on free electrons exactly balances gravity, setting the natural ceiling above which a thin disk cannot remain in steady-state. For a

BH of mass M it is given by

$$\dot{M}_{\text{Edd}} = \frac{4\pi G M m_p}{c\eta\sigma_t} \quad (6)$$

where m_p is the proton mass, σ_t is the Thomson scattering cross-section and $\eta \equiv 0.1$ is the radiative efficiency. We are then able to retain information about the relative accretion rates of the two BHs, we set the binary accretion rate $\dot{M}_b \equiv \dot{M}_1 + \dot{M}_2$ equal to $\gamma \dot{M}_{\text{Edd}}$ (with γ an arbitrary scale factor evaluated at the total binary mass), and convert the accretion rate of each individual BH to units of $\dot{M}_{\text{Edd}}$ its own Eddington units.

In Fig. 11 we set the accretion rate of the binary to be $1\dot{M}_{\text{Edd}}$ assuming a $10^7 M_\odot$ binary and plot the accretion rate for both BHs normalized to their respective Eddington accretion rates. The horizontal gray dashed line is at $1\dot{M}_{\text{Edd}}$ to represent the accretion rate above which jets are likely to launch. The red lines represent the accretion rate of the primary, and the black lines represent the accretion rates of the secondary. The background color of the panel is associated with different jet-behaviors: purple for dual jets, blue for a single jet, green for flickering jets. The time-slice displayed is arbitrary and serves to merely highlight the accretion behavior.

What first strikes us in Fig. 11 is the wide variety of difference in magnitude between the two BHs' accretion rates. Since the Eddington accretion rate scales linearly with the BH mass we expect that the accretion rates of the secondary to be increased greatly when normalized to Eddington units. This is evidenced by the nearly 2 order of magnitude difference between the secondary and primary at $q_b = 0.1$. Further, we also notice that the behavior of the individual accretion rates of the black holes are quite varied, as the $\lambda(t)$ results suggested. Aside from the differences in whether the BHs' accretion rates are stable or not, the profile of the accretion rate itself is varied. Some binaries experience accretion rates that are close to sinusoidal (e.g. $e_b = 0.6, 0.8$), others seem to closer resemble square-waves (e.g. $q_b = 1, e_b = 0.3, e_b = 0.5$; e.g. $e_b = 0.3$ and 0.5 at $q_b = 1$), others yet have quite sharp breaks that evade simple characterizations (e.g. $e_b = 0.1, q_b = 0.7$; e.g. $(e_b, q_b) = (0.1, 0.7)$). Further, we note that the accretion rate of one BH is not always simply the accretion rate of the other with a different baseline and $\pi/2$ phase shift. Rather, they can take on notably different profiles from each other, resulting, at times, in both BHs experiencing a local peak in accretion rate, but because of different accretion rate amplitudes result in a peak in λ . It is exactly this zoo of individual BH accretion rates, and the way in which they compare to each other, that yield an interesting assortment of jet-launching behaviors.

In Fig. 11 we delineate three broad regimes: a) binaries where one BH launches a jet (blue), b) binaries where both BHs coincidentally launch jets in a sustained and repeated fashion (purple), c) binaries where both BHs launch jets in a successive, alternating fashion (green). We describe these jet-behavior regimes as single jets, dual jets, and flickering jets, respectively. We assigned jet-regimes by determining whether the accretion rate of each BH surpassed a threshold value of $1.1\dot{M}_{\text{Edd}}$ for greater than 50τ . a threshold chosen modestly above unity to allow for the disk thickness to inflate enough to support the funnel collimation discussed above — and whether those instances were temporally coincident for greater than 50τ . The 50τ duration reflects the physical timescale over which jets are expected to launch and quench: jet activity follows the inner-disk dynamical time, $t_{\text{dyn}} \sim \Omega_K^{-1}$, which for our 2D setup is of order a few binary orbital periods, so threshold-crossings sustained for $\geq 50\tau$ correspond to many dynamical times of stable launching.

We note that the majority of simulations at the assumed binary accretion rate of $1\dot{M}_{\text{Edd}}$ are single-jet systems, where most of them have

low q_b due to the highly unequal Eddington-normalized accretion rates of those systems.

The flickering jet systems are clustered at higher q_b and e_b . Due to large λ amplitudes, such systems are able to exist up to $q_b = 0.5$. It is important to note that since these flickering jets are dependent on large λ oscillations they are unique to $e_b \neq 0$ binaries.

Finally, we have a few dual-jet systems. For those at $e_b = 0$ both BHs accrete at nearly the same, stable rate and thus both launch sustained jets. However, we also note that we have dual jets for binaries whose $\lambda \neq 1$. We note that the dual-jet systems at $e_b = 0.1$ and $e_b = 0.8$ show both BHs having coincident, local peaks in accretion rates, meaning that the binary sustainably and periodically launches two jets.

While dual jets from BBH systems have been suggested before (Palenzuela et al. 2010; Qian et al. 2019) and have even been simulated (Gutiérrez et al. 2024; Ressler et al. 2025) demonstrated numerically in a range of GR-MHD setups (Gutiérrez et al. 2024; Ressler et al. 2025; Ruiz et al. 2023; Most & Wang 2024;), we believe that we are the first to report evidence suggesting a flickering-jet behavior. An entirely new regime, it not only provides a unique observational determination of eccentric binaries but it is also — alternating, rather than coincident, jet activity from the two BHs. We emphasize that the prior GR-MHD work has typically begun from configurations in which jet conditions are already satisfied at both BHs (e.g., a magnetically arrested circumbinary flow) and asks how the resulting jets evolve through inspiral. Our framing is complementary: we ask, given a hydrodynamic accretion-rate time series, when each BH's accretion crosses the threshold required for jet activity in the first place. The flickering regime is unique to $e_b \neq 0$ binaries with large λ amplitudes, and provides a uniquely distinct kind-of-feedback mechanism that could be further explored in cosmological simulations.

Fig. 11 and its discussion assumed a binary accretion rate of $1\dot{M}_{\text{Edd}}$. However, such this assumption only controls the baseline of the binary accretion rate. While a different assumed binary accretion rate would not change the profiles nor the relative accretion rates of the components, it would affect the absolute amplitude — changing its amplitude — changing the jet-launching behavior. To make this dependence explicit, Fig. 13 shows the jet-regime classification across (e_b, q_b) for three representative binary accretion rates spanning the dynamic range over which all three regimes coexist. As $\dot{M}_b$ rises from $3\dot{M}_{\text{Edd}}$ to $10\dot{M}_{\text{Edd}}$, single jets recede toward low q_b and low e_b , dual jets fill in across most of parameter space, and flickering jets concentrate in the upper-right corner of high q_b and high e_b where the two BHs' accretion rates straddle the threshold and cross it alternately. At rates well below this range no jets launch anywhere in (e_b, q_b) space; at rates well above, dual jets dominate the entire parameter space. The fiducial diagram in Fig. 11 represents one slice of this continuous family.

Fig. 12 displays $\tilde{\lambda}$, which can be written as

$$\tilde{\lambda} \equiv \frac{\langle \max(\dot{M}_1, \dot{M}_2) \rangle}{\langle \min(\dot{M}_1, \dot{M}_2) \rangle} \quad (7)$$

where max and min are element-wise functions and the accretion rates are provided in Eddington units.

The result $\tilde{\lambda}$ is dimensionless and provides the ratio of the maximum and minimum Eddington-normalized accretion rates of the binary components. Given the time-variability of λ in Table 1 and a binary accretion rate, $\tilde{\lambda}$ can then be easily used to determine whether one or both BHs launch a jet, and whether they are temporally coin-

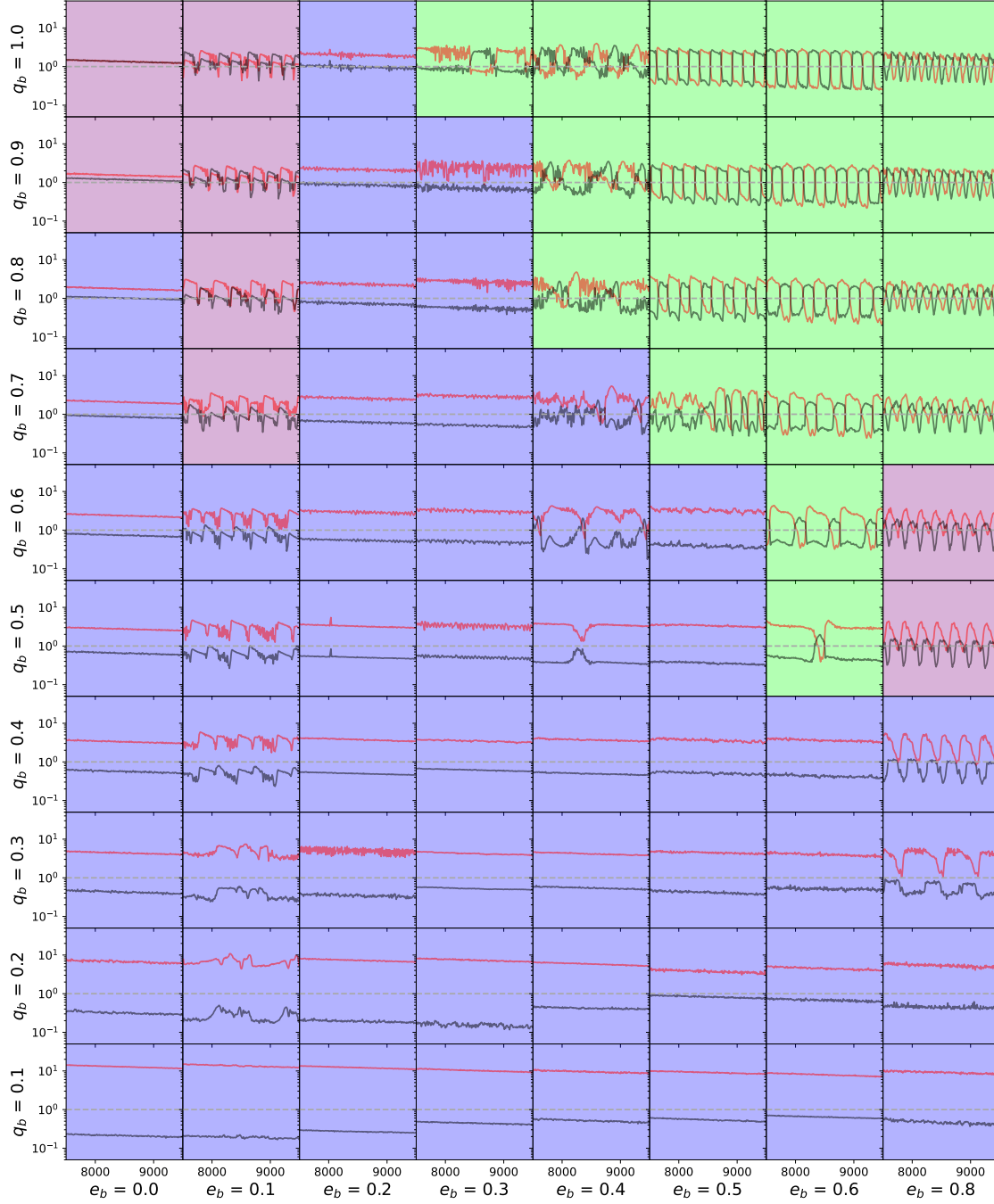


Figure 11. Depiction of Eddington-normalized accretion rates for our simulation suite, with panel backgrounds colored by jet-regime. The panel structure is the same as that of Fig. 1, with time on the x-axis of each panel, accretion rate, in Eddington units, on the y-axis. We display the accretion rate of the primary (red), the secondary (black) and the jet-launching threshold of $1M_{\text{Edd}}$ (horizontal dashed gray). Blue background colors indicate single jet regimes, purple indicates dual-jet regimes, and green indicates flickering jet regimes.

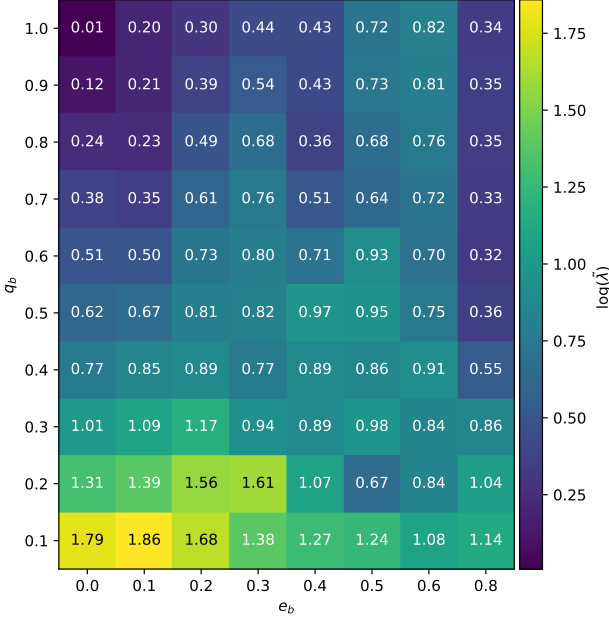


Figure 12. $\tilde{\lambda}$ —The Eddington-normalized accretion-rate gauge $\tilde{\lambda} \equiv (\max(\dot{M}_1, \dot{M}_2)) / (\min(\dot{M}_1, \dot{M}_2))$ for our simulation suite, in $\dot{M}_{\text{Edd}}$ units. The x-axis is the binary eccentricity e_b , and the y-axis is q_b , and the binary mass-ratio q_b ; cell colors indicate $\log \tilde{\lambda}$. Combined with an assumed binary accretion rate $\dot{M}_b$, $\tilde{\lambda}$ converts directly into the value Eddington-normalized accretion rate of $\log(\tilde{\lambda})$ each BH, allowing the jet-launching regime (single, dual, or flickering — see Fig. 11) to be predicted from (e_b, q_b) alone for a given $\dot{M}_b$.

cident. Namely, one can use

$$\dot{M}_1 = \frac{\dot{M}_b}{\tilde{\lambda} + 1} \quad (8)$$

to find the accretion rate of the primary based on the scaling of λ and $\dot{M}_b$. With Fig. 12 in hand we are able to gain insight into the jet behavior of any proposed BBH system as a function of q_b and e_b . Given the clustering of jet-behavior in q_b - e_b parameter space, determination of whether a system has sustained dual jets, flickering jets, or a single jet could greatly constrain the orbital parameters of the system.

“Flickering jets” are a unique EM signature for BBHs. In order to use them to find BBH systems, we must ensure they “flicker”—i.e. switch which BH is preferentially accreting—on a human timescale. In the following we compute and place constraints on the time to “flicker.”

Firstly, we require that the time to flicker τ_f be fewer than 10 years in the observer’s rest-frame, so that a few cycles could be possible in a human lifetime. For simplicity, we assume the flickering period is 300τ . We also require that the binary not merge in less than 100 years, in order to ensure that these systems are common enough.

In Fig. 14 we display the region of parameter space that satisfies the above time constraints (within black lines) for various binary masses at various redshifts. The x-axis of each panel is the eccentricity e_b , the y-axis is the a_b in Schwarzschild radii, and τ_f is reported in the observer frame. The gray lines represent the change in eccentricity and semi-major axis for the binary — due to 10 orbits worth of GW radiation, computed via Peters (1964).

Fig. 14 shows that $10^5 M_\odot$ and $10^6 M_\odot$ systems both display consid-

erable regions of parameter-space that satisfy the constraints necessary to observe the flicker, up to $z = 10$. We find that the region which satisfies these constraints shrinks to zero by $z = 15$ and $z = 100$ for the $10^6 M_\odot$ and $10^5 M_\odot$ systems, respectively. This provides encouraging evidence for our ability to observe these flickering jet systems.

In addition to observing a flicker occur, we note that jets are extended emission sources and thereby provide us an ability to observe a past flicker. If we could determine a geometric separation in the structure of a helical jet, this could indicate that the emission is from a binary that flickered in the past.

4.2 Unequal Mass Sources

In addition to the variety of jet launching behavior and their effects on galaxy feedback, the evolution of the mass ratio of these SMBBHs is also a key observational signature. We explore whether our findings from Fig. 10 suggesting that some binaries do not evolve towards equal mass in fact lead to detectable $q \neq 1$ BBH systems.

In addition to its mass-ratio, the eccentricity and semi-major axis of the binary also change as functions of q_b and e_b due to the CBD, we represent these as $\dot{e}_{\text{gas}}(q_b, e_b)$ and $\dot{a}_{\text{gas}}(q_b, e_b)$, respectively. The values for $\dot{a}_{\text{gas}}(q_b, e_b)$, and $\dot{e}_{\text{gas}}(q_b, e_b)$ were also calculated for this full simulation suite and are reported in Figure 2 and 3 of Siwek et al. (2023b), respectively. Since our simulation suite is discretized in e_b and q_b space, we define $\dot{e}_{\text{gas}}(q_b, e_b)$, $\dot{a}_{\text{gas}}(q_b, e_b)$ and $\dot{q}(q_b, e_b)$ as sets of piecewise functions with constant values that are centered at the binary parameter values and extend to the midpoint between it and its neighboring, sequential, binary parameter value. For example,

$$\dot{q} = 1.041 \begin{cases} 0.15 \leq q_b < 0.25 \\ 0.15 \leq e_b < 0.25 \end{cases} \quad (9)$$

However, in addition to CBD effects, the GWs radiated from the system also circularize and shrink the binary; we include these effects through the terms—standard quadrupole-formula expressions (Peters 1964)

$$\dot{a}_{\text{GW}} = \frac{-64G^3 M^3 q f(e)}{5c^5 a^3 (1+q)^2} \quad (10)$$

and

$$\dot{e}_{\text{GW}} = -e \frac{304G^3 M^3 q \left(1 + \frac{121}{304} e^2\right)}{15c^5 a^4 (1-e^2)^{5/2} (1+q)^2} \quad (11)$$

Thus, we are left with the set of coupled differential equations

$$\dot{a}(q_b, e_b) = \dot{a}_{\text{gas}}(q_b, e_b) + \dot{a}_{\text{GW}}(q_b, e_b, a_b) \quad (12)$$

$$\dot{e}(q_b, e_b) = \dot{e}_{\text{gas}}(q_b, e_b) + \dot{e}_{\text{GW}}(q_b, e_b, a_b) \quad (13)$$

$$\dot{q}(q_b, e_b) = \dot{q}_{\text{gas}}(q_b, e_b) \quad (14)$$

$$(15)$$

which we numerically integrate given an initial a_0 , q_0 , and e_0 .

Per Fig. 10, we solve the initial value problem for two equal-mass binaries with $e_b = 0.2$ and $e_b = 0.3$, $q_b = 0.2$ and $q_b = 0.3$, integrating forward for $10^4 \tau$ with $a_0 = 10^3 R_S$. The results are displayed in Fig. 15. We assume a binary accretion rate $\dot{M}_b = 100 \dot{M}_{\text{Edd}}$ throughout. This assumption sets the timescale over which the gas-driven equilibrium is reached but does not change the qualitative endpoint of the evolution over our integration interval. We retain time as the x-axis in Fig. 15 (rather than semi-major axis) because the relaxation to the gas-driven equilibrium is more naturally read in time units.

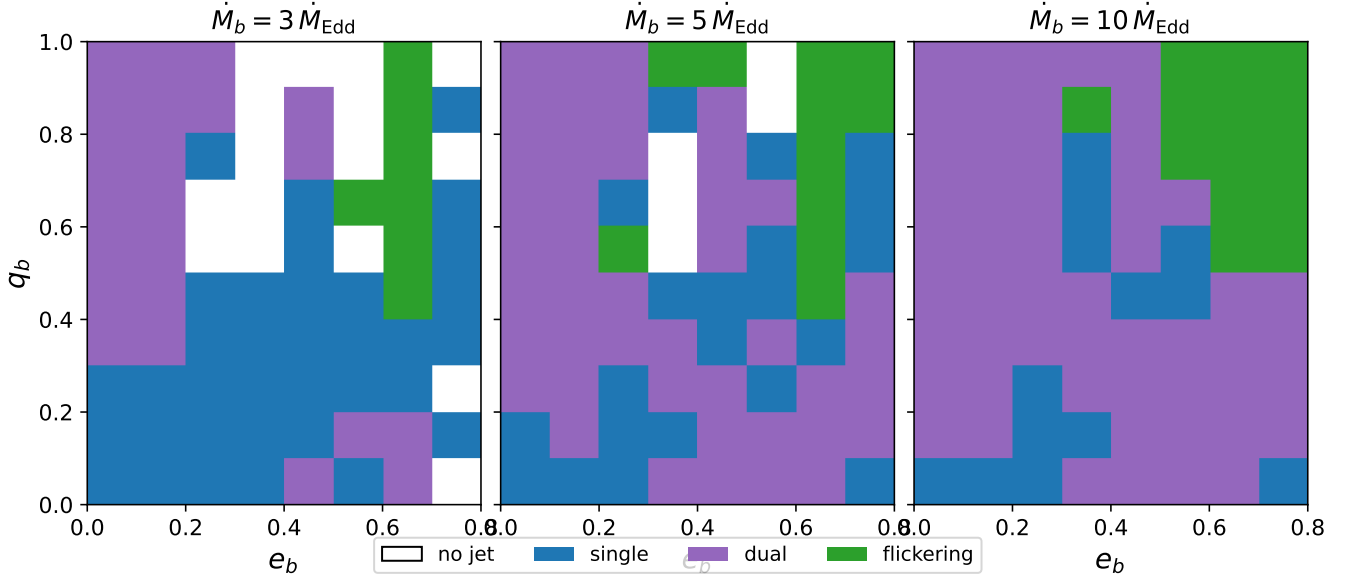


Figure 13. Jet-regime classification as a function of (e_b, q_b) for three values of the binary accretion rate $\dot{M}_b \in \{3, 5, 10\} \dot{M}_{\text{Edd}}$. Each cell is colored by the regime (white: no jet; blue: single; purple: dual; green: flickering) using the same threshold and duration criteria as Fig. 11 ($\dot{M}_i > 1.1 \dot{M}_{\text{Edd},i}$ sustained for $\geq 50 \tau_b$). The three values bracket the range over which all three regimes coexist: at $\dot{M}_b = 3$ singles dominate with dual at the low- e_b , low- q_b edge and flickering along $e_b \approx 0.6$; at $\dot{M}_b = 5$ duals fill most of the grid and flickering migrates upward in q_b ; at $\dot{M}_b = 10$ duals dominate and flickering occupies the high- q_b , high- e_b corner.

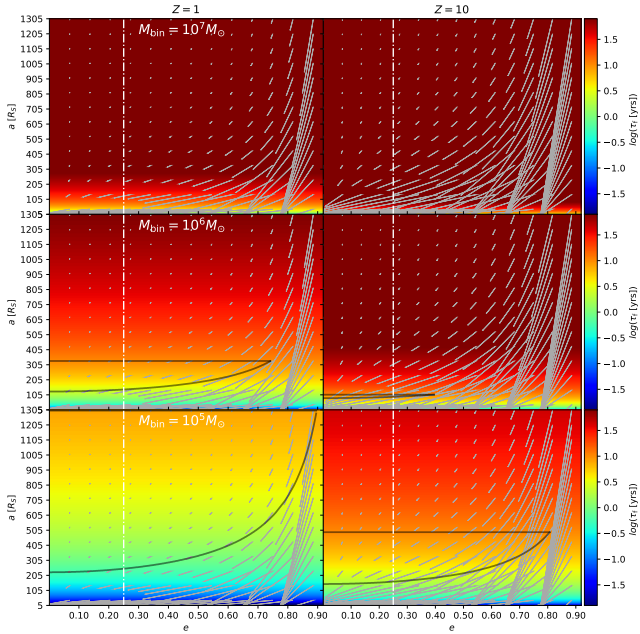


Figure 14. The region of parameter space that satisfies the above time constraints (within black lines) for various binary masses at various redshifts. The x-axis of each panel is the eccentricity e_b , the y-axis is the a_b in Schwarzschild radii, and τ_f is reported in the observer frame. The gray lines represent the net effect of 10 orbits of GW radiation on the eccentricity and semi-major axis of each BBH.

In Fig. 15 we display the evolution of the two binary's $e(t)$ and $q(t)$ in the upper and lower panels, respectively. ~~Firstly, our numerical integration confirms the expectation that the binary evolves towards the expected $e_b = 0.45$ equilibrium eccentricity (Duffell et al. 2020; Zrake et al. 2021; Siwek et al. 2023b, 2024).~~ ~~More interestingly, we~~ Our numerical integration is performed at $a_0 = 10^3 R_S$, where the gas-driven terms $\dot{a}_{\text{gas}}$ and $\dot{e}_{\text{gas}}$ dominate over the GW terms (Peters 1964), and the binary settles toward the gas-driven equilibrium $e \approx 0.45$ reported by Duffell et al. (2020); Zrake et al. (2021); Siwek et al. (2023b, 2024). We find that the two binaries initialized at $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ do indeed $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ both evolve away from $q_b = 1$, $q_b = 1$, reaching $(e, q) = (0.45, 0.991)$ and $(0.45, 0.998)$ respectively, before further evolution.

These plateau values are not the final pre-merger state. As the binary inspirals and a shrinks, the GW timescale $\tau_{\text{GW}} \propto a^4$ (Peters 1964) eventually becomes shorter than the gas-driven evolution timescale, and GW radiation begins to dominate the orbital dynamics. Crucially, the Peters quadrupole formulas conserve the mass ratio q at leading order: GW radiation drives $a \rightarrow 0$ and ~~have end-values~~ of $(0.45, 0.991)$ $e \rightarrow 0$ but does not directly change q . The mass-ratio offset established during the gas-dominated phase is therefore preserved through inspiral rather than driven back toward unity. We have verified this by extending the integration of equation 12 for the $(e_b, q_b) = (0.2, 1.0)$ and $(0.45, 0.998)$, respectively. These results suggest that BBHs, which at some point in their evolution, have the parameters $(0.2, 1.0)$ $(0.3, 1.0)$ cases from $a_0 = 10^3 R_S$ down to $a = 5 R_S$: GW circularizes e to ≈ 0 before LISA-band entry, but q remains essentially constant at the gas-phase plateau. A scan over initial separation finds that the bump effect

⁷ We note that

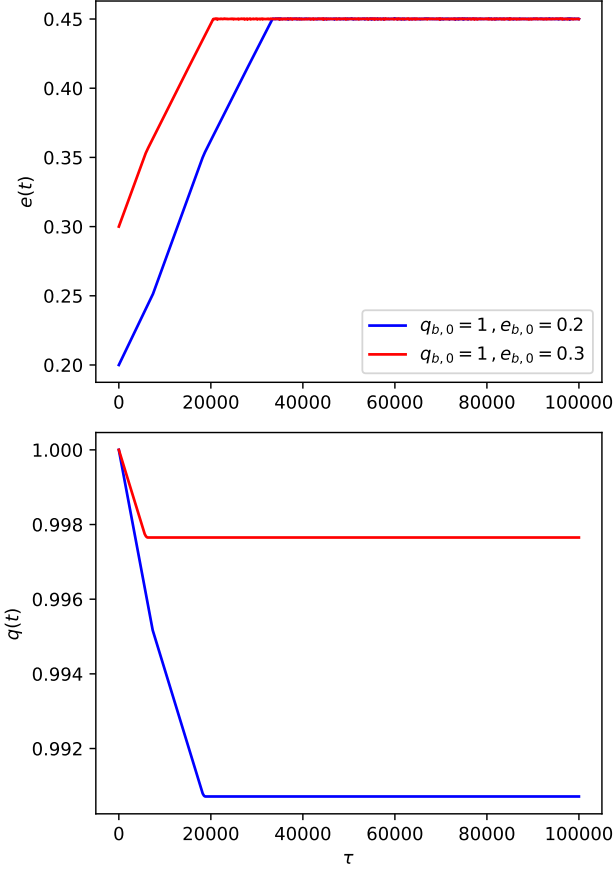


Figure 15. Evolution of binaries away from equal-mass. We depict the result of our numerical integration of equation 12 for $e_{b,0}, q_{b,0} = (0.2, 1.0)$ (blue) and $(0.3, 1.0)$ (red). The y-axes are $e(t)$ (upper) and $q(t)$ (lower) of the binary and the x-axes are time in units of each binary’s orbital period.

activates by $a_0 \gtrsim 250R_S$ and reaches its asymptotic value by $a_0 \gtrsim 400R_S$; for an $M_{\text{tot}} = 10^7 M_\odot$ binary entering inspiral with $\dot{M}_b = 100 \dot{M}_{\text{Edd}}$, the asymptotic mass-ratio offsets at LISA-band entry are $\Delta q \approx 0.014$ and $\Delta q \approx 0.008$ for the two cases, respectively.

Two caveats apply to this prediction. First, the simulation suite that supplies $\dot{q}_{\text{gas}}$ is gridded at $\Delta q_b = 0.1$ resolution. Once q drifts from 1.0 to ~ 0.99 during inspiral, our integration samples the $q_b = 0.9$ row of the lookup table, which corresponds to simulations of binaries that had $q_b = 0.9$ throughout — not binaries that started at $q = 1$ and drifted. We do not have direct simulation data in $q \in (0.95, 1.0)$. Second, the cavity geometry depends on the binary’s history: a binary that started at $q = 1$ has a CBD locked toward the spatially-defined “secondary” (an apocenter convention), whereas a $q_b = 0.9$ simulation has a CBD locked toward the lower-mass BH from the start. As q drifts past unity, the cavity must reorient, which we argued in Section 3.2 proceeds on the disk’s apsidal precession timescale — a quantity we have not compared to the gas-driven q -evolution timescale. If the cavity reorients faster than q evolves, the lookup remains applicable; if slower, the relevant $\dot{q}_{\text{gas}}$ during inspiral could differ from what our integration assumes. Resolving these caveats would require new simulations at finer q -grid resolution and live-binary runs through a sustained $\dot{q} \neq 0$ phase, both of which we leave to future work.

Subject to these caveats, we predict that LISA-detectable

binaries that passed through $(e_b, q_b) = (0.2, 1.0)$ or $(0.3, 1.0)$ have equilibrium mass-ratios that are not equal to one. LISA is expected to have a during the gas-dominated phase of their formation should appear at $q \approx 0.99$ rather than at $q = 1$, producing a relative dearth of strictly $q = 1$ systems in the LISA mass-ratio distribution. Whether this leaves a measurable population-level signature depends on three additional open questions: (i) the fraction of MBHB progenitors that pass through the relevant region of (e_b, q_b) parameter space and the duty cycle of the $q \neq 1$ phase across the population, (ii) LISA’s expected $\sim 0.5\%$ precision in q_b of up to 0.5% (Mangiagli et al. 2020). Thus, while the deviation from $q_b = 1$ for the end state of $(0.3, 1.0)$ binaries may not be detectable by LISA, that of $(0.2, 1.0)$ will be within detection limits. Thus, given enough LISAdetections, a number-density bump at $q < 1$ could suggest the history of the binaries, indicating it passed through $(q_b = 1, e_b = 0.2)q$ (Mangiagli et al. 2020) relative to the $\Delta q \sim 0.01$ deviation, and (iii) the expected $\mathcal{O}(100)$ MBHB events over the LISA mission. Confirming this prediction would require both a careful population synthesis accounting for the duty cycle of the $q \neq 1$ phase and a forward-modelling of LISA’s detection sensitivity for a near-equal-mass population.

5 SUMMARY AND CONCLUSIONS

This paper has provided the most extensive report on the preferential accretion $\lambda \equiv \frac{\dot{M}_2}{\dot{M}_1}$ and mass ratio rate of change $\dot{q}$ for SMBBHs embedded in CBDs. In addition to calculating quantities, we provide novel insight into the behavior of these values over time and their dependence on q_b and e_b . We also conduct a preliminary investigation into how the CBD regulates preferential accretion. We summarize our key findings below.

- (i) Across q_b and e_b , $\lambda(t)$ can be split into time-varying and time-stable regimes, broadly mirroring the split of the CBD into precessing and locked states.
- (ii) The average preferential accretion value $\langle \lambda \rangle$ exhibits a weak positive correlation with both e_b and q_b . The standard deviation of λ , σ_λ , exhibits a weak negative correlation with both e_b and q_b .
- (iii) $\langle \lambda \rangle$ is positively correlated with more eccentric CBDs.
- (iv) We find strong evidence that while the CBD precession and across precessing systems in our suite, the CBD apsidal precession period and the $\lambda(t)$ oscillation have the same period are equal. However, these time-series are not in phase, ruling out a simple causal interpretation in terms of cavity-wall distance.
- (v) We do not find evidence that a BH must be closer to the CBD cavity than its counterpart to accrete preferentially at a higher rate.
- (vi) Normalized to Eddington accretion rates, $\lambda(t)$ results in disparate accretion regimes for each BH in the binary—leading to unique jet-launching regimes. We delineate binaries that are likely to launch a sustained jet from one BH (single-jet), from each BH (dual-jet), or alternate in which BH launches a jet (flickering-jet).
- (vii) We find that $e_b, q_b = (0.2, 1.0)$ and $(0.3, 1.0)$ binaries do not have $q_b = 1.0$ steady-states and Equal-mass binaries at $e_b = 0.2$ and 0.3 are not long-lived equilibrium states: during the gas-dominated phase of evolution they preferentially accrete away from equal-mass. The steady-state of a $(0.2, 1.0)$ system will be detectable by LISA as having unequal-mass, reaching $q \approx 0.991$ and 0.998 respectively. Because GW radiation conserves the mass ratio at leading order, this offset is preserved through inspiral. We therefore predict a relative dearth of strictly $q = 1$ systems in the LISA

mass-ratio distribution; whether this is detectable depends on population statistics and forward-modelling we leave to future work.

While our work has shed light on novel aspects of this system, we are still studying setups with simplified physics, namely two-dimensional hydrodynamic simulations. In future work we hope to expand this study to more physically motivated simulation setups, accounting for effects that were ignored in this study. Namely, we hope to detail λ and $\dot{q}$ for radiation MHD simulations in three dimensions. Further, we would like to also develop GRMHD simulations for SMBBHs in order to better simulate jet-launching and test our findings on different jet regimes. In addition to developing more robust simulations, we would also like to run live-binary simulations for the $e_b, q_b = (0.2, 1.0)$ ($e_b, q_b = (0.2, 1.0)$) and $(0.3, 1.0)$ binary systems to not only better extrapolate how the binary evolves away from unity but to also understand how the disk reacts to the change in position of the primary and secondary. Finally, we will further investigate the mechanism behind preferential accretion, with a forthcoming analysis on the hydrodynamics of the gas within the cavity.

We conclude by noting that ~~while~~ the mechanism behind preferential accretion is more complex than previously thought, ~~it is still intimately tied to the CBD and can have profound observational consequences for SMBBH systems, such as jet-launching and LISA population studies.~~ Although our results are consistent with the CBD playing a regulating role, the most concrete observational handle is the flickering-jet regime; the population-level LISA signature is intriguing but conditional on the binary formation distribution. We encourage further study into ~~the preferential accretion mechanism and its consequences, both.~~

ACKNOWLEDGEMENTS

The authors thank the anonymous referees for helpful comments. ZH acknowledges support from NASA ATP grant [INSERT NUMBER] and LISA Preparatory Science grant [INSERT NUMBER]. We acknowledge computing resources from Columbia University's Shared Research Computing Facility, in particular the Ginsburg HPC cluster.

DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author.

6

1.0 ----- 0.9 ----- 0.8 ----- 0.7 -----
----- 0.6 ----- 0.5 ----- 0.4 -----
----- 0.3 ----- 0.2 ----- 0.1 -----
----- q_b/e_b 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.8

Reproduction of Table 1 from DeLaurentiis & Rafikov (2025), showing whether the CBD about a binary of given q_b and e_b is locked (L; red) or precessing (P; blue). The CBD behavior is determined by evaluating the time-series of the eccentricity vector at $a \leq 5a_b$.

Reproduction of Figure 18 from DeLaurentiis & Rafikov (2025), displaying the semi-major axis and eccentricity of the cavities of our simulations with varying binary eccentricity and mass ratio. The cavity is characterized by the $m = 1$ Fourier mode of the CBD cavity

REFERENCES

- Artymowicz P., 1983, *Postepy Astronomii Krakow*, **31**, 19
- Barnes J. E., Hernquist L., 1992, *ARA&A*, **30**, 705
- Begelman M. C., Blandford R. D., Rees M. J., 1980, *Nature*, **287**, 307
- Calcino J., Dempsey A. M., Dittmann A. J., Li H., 2023, *arXiv e-prints*, p. [arXiv:2311.13727](#)
- D’Orazio D. J., Duffell P. C., 2021, *Astrophysical Journal Letters*, **914**, L21
- D’Orazio D. J., Haiman Z., MacFadyen A., 2013, *Monthly Notices of the Royal Astronomical Society*, **436**, 2997
- D’Orazio D. J., Duffell P. C., Tiede C., 2024, *ApJ*, **977**, 244
- DeLaurentiis S., Rafikov R., 2025, in preparation
- DeLaurentiis S., Haiman Z., Westernacher-Schneider J. R., Krauth L. M., Davelaar J., Zrake J., MacFadyen A., 2025, *ApJ*, **980**, 55
- Dittmann A. J., Ryan G., 2021, *ApJ*, **921**, 71
- Dittmann A. J., Dempsey A. M., Li H., 2023, *arXiv e-prints*, p. [arXiv:2310.03832](#)
- Duffell P. C., D’Orazio D., Derdzinski A., Haiman Z., MacFadyen A., Rosen A. L., Zrake J., 2020, *ApJ*, **901**, 25
- Ennoggi L., Combi L., Campanelli M., et al., 2025, *arXiv e-prints*
- Farris B. D., Duffell P., MacFadyen A. I., Haiman Z., 2014, *Astrophysical Journal*, **783**, 134
- Goodchild S., Ogilvie G., 2006, *Monthly Notices of the Royal Astronomical Society*, **368**, 1123–1131
- Günther R., Kley W., 2002, *A&A*, **387**, 550
- Guolo M., Ruschel-Dutra D., Grupe D., Peterson B. M., Storch-Bergmann T., Schimoia J., Nemmen R., Robinson A., 2021, *MNRAS*, **508**, 144
- Gutiérrez E. M., Combi L., Romero G. E., Campanelli M., 2024, *MNRAS*, **532**, 506
- Kley W., Papaloizou J. C. B., Ogilvie G. I., 2008, *Astronomy and Astrophysics*, **487**, 671
- Lee W.-K., Dempsey A. M., Lithwick Y., 2019, *The Astrophysical Journal Letters*, **882**, L11
- Lubow S. H., 1991, *Astrophysical Journal*, **381**, 259
- Lubow S. H., 2022, *Monthly Notices of the Royal Astronomical Society*, **516**, 5446–5453
- MacFadyen A. I., Milosavljević M., 2008, *ApJ*, **672**, 83
- Mangiagli A., et al., 2020, *Phys. Rev. D*, **102**, 084056
- Miranda R., Muñoz D. J., Lai D., 2017, *Monthly Notices of the Royal Astronomical Society*, **466**, 1170
- Moriwaki K., Nakagawa Y., 2004, *Astrophysical Journal*, **609**, 1065
- Most E. R., Wang H.-Y., 2024, *ApJ*, **973**, L19
- Muñoz D. J., Lithwick Y., 2020, *Astrophysical Journal*, **905**, 106
- Muñoz D. J., Miranda R., Lai D., 2019, *ApJ*, **871**, 84
- Nelson R. P., 2003, *Monthly Notices of the Royal Astronomical Society*, **345**, 233
- Ochi Y., Sugimoto K., Hanawa T., 2005, *ApJ*, **623**, 922
- Paardekooper S. J., Thébault P., Mellema G., 2008, *Monthly Notices of the Royal Astronomical Society*, **386**, 973
- Palenzuela C., Lehner L., Liebling S. L., 2010, *Science*, **329**, 927
- Peters P. C., 1964, *Physical Review*, **136**, 1224
- Qian S. J., Britzen S., Krichbaum T. P., Witzel A., 2019, *A&A*, **621**, A11
- Rafikov R. R., 2016, *ApJ*, **827**, 111
- Ragusa E., Lynch E., Laibe G., Longarini C., Ceppi S., 2024, *Astronomy and Astrophysics*, **686**, A264
- Ressler S. M., Combi L., Ripperda B., Most E. R., 2025, *ApJ*, **979**, L24
- Ruiz M., Tsokaros A., Shapiro S. L., 2023, *Phys. Rev. D*, **107**, 103025
- Shi J.-M., Krolik J. H., Lubow S. H., Hawley J. F., 2012, *ApJ*, **749**, 118
- Siwek M., Weinberger R., Muñoz D. J., Hernquist L., 2023a, *Monthly Notices of the Royal Astronomical Society*, **518**, 5059
- Siwek M., Weinberger R., Hernquist L., 2023b, *Monthly Notices of the Royal Astronomical Society*, **522**, 2707
- Siwek M., Kelley L. Z., Hernquist L., 2024, *MNRAS*, **534**, 2609
- Springel V., 2010, *MNRAS*, **401**, 791
- Statler T. S., 2001, *The Astronomical Journal*, **122**, 2257–2266
- Teyssandier J., Ogilvie G. I., 2016, *Monthly Notices of the Royal Astronomical Society*, **458**, 3221–3247
- Thun D., Kley W., Picogna G., 2017, *Astronomy and Astrophysics*, **604**, A102

- Tiede C., Zrake J., MacFadyen A., Haiman Z., 2020, [Astrophysical Journal](#), **900**, 43
- Tiede C., Zrake J., MacFadyen A., Haiman Z., 2022, [ApJ](#), **932**, 24
- Tiede C., D’Orazio D. J., Zwick L., Duffell P. C., 2024, [Astrophysical Journal](#), **964**, 46
- Westernacher-Schneider J. R., Zrake J., MacFadyen A., Haiman Z., 2022, [Phys. Rev. D](#), **106**, 103010
- White S. D. M., Rees M. J., 1978, [MNRAS](#), **183**, 341
- Whitehurst R., 1994, [Monthly Notices of the Royal Astronomical Society](#), **266**, 35
- Young M. D., Baird J. T., Clarke C. J., 2015, [MNRAS](#), **447**, 2907
- Zrake J., Tiede C., MacFadyen A., Haiman Z., 2021, [Astrophysical Journal Letters](#), **909**, L13

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